Jackson Harris

(224) 655-8146 | jkharris2023@gmail.com

Aspiring astrophysicist applying advanced mathematics and data analysis to questions about galactic structure and particle physics.

EDUCATION

Barrington High School, Barrington, Illinois (Class of 2027)
Weighted GPA: 4.524 **SAT:** 1560 (790 Math, 760 Reading and Writing)
AP Courses and Scores: Human Geography (5), Computer Science Principles (5), United States History (4), Computer Science A (5), Chemistry (5), Calculus BC (5) with AB Subscore (5), United States Government and Politics (5), Physics C: Mechanics (5), Physics C: Electricity and Magnetism (5), English Language and Composition (5)
Advanced Classes Completed: Multivariable Calculus, Linear Algebra
Advanced Classes in Progress: AP Biology, AP Psychology, AP Literature and Composition, AP Microeconomics, AP Macroeconomics, AP Spanish, Computer Science Advanced Topics
College Courses: Calculus III at UIUC, Linear Algebra at UIUC

SCIENTIFIC WORKS

Astrophysics Research Paper **May 2026 – Present**

- Led an in-depth work on the identification of stellar clusters in the Milky Way using unsupervised machine learning algorithms to calculate the Oort constants and a rotation curve. This work includes an in-depth error analysis and aims to push the bounds of accuracy. The work was submitted to the Experimental Astronomy Journal of Springer Nature.

Award Winning Beamline Experiment Paper for CERN **December 2025 – March 2026**

- Wrote a consolation winning experiment paper which proposes the studies of different muon beam dump materials. This paper is motivated by findings from the use of GEANT4 on experiments using novel metallic alloys.

Portfolio Website and active Blog **August 2024 – Present**

- Maintains a portfolio website that includes a blog with 20+ articles and information on myself.
- https://jackson-harris-247.github.io/Portfolio/#

Published Projects on GitHub in contribution to the open source community **December 2022 – Present**

- Multiple personal projects and contributions on GitHub in fields I simply find interesting such as computer simulations and mathematics.

Leadership, Service, and Work

Lead Role in my Community's Largest Performance of the Year **February – March 2026**

- Selected for the lead role of Jesus in Holy Family Church's Passion Play, demonstrating leadership by guiding fellow cast members through months of preparation and multiple live Easter performances to over 4,000 in person viewers.

Swoboda Deaf Center Project Lead **May 2026 – Present**

- Leading the Restoration of the Swoboda Deaf Center and building the only Deaf Museum in the city of Chicago, with accessible digital interface and archival resources. Raised $9,500 with personal outreach and $5,000 in matching pledges for a total of $14,500.

President of Physics Club **June 2026 – Present**

- The current leader of the physics club, with organized meetings, extracurricular events, and physics competitions. The club exists with the goal to spread interest in the field especially among underrepresented groups. Raised club participation by 100%.

Local Track and Field Coach **August 2024 – Present**

- Volunteer boys and girls track and field coach for 35+ middle schoolers at Holy Family Catholic Academy. Educated athletes on discus and shot put technique with multiple athletes reaching the state level.

Youth Football Referee **August 2025 – Present**

- Ensured fair and safe play across 16 games as a paid head referee for boys and girls flag football across age levels.

Home Renovation **August 2024 – Present**

- **G**eneral contractor for McMahon Investco which led to the renovation of two homes to livability. Performed carpentry work with floor and wall repair, painting, and pipework.

Meal Planning and Preparation for Saint Leonard's House Chicago **February 2025**

- Organized the preparation, delivery, and serving of food in large quantities to the recently incarcerated at Saint Leonard's.

Youth Faith Group Based Community Service **August 2024 – Present**

- Participate in faith based shopping for Christmas gifts for underprivileged families and feeding the homeless.

AWARDS & HONORS

State Awards for Math team **March 2025**

- ICTM 2024-2025 1st place finish.

CERN BL4S Top 15 Winner **June 2026**

- As a part of the physics club our team won a top 15 prize out of the 712 teams that participated from 89 distinct countries. The prize includes a telescope signed by multiple Nobel laureates and the General Director of CERN.

AIME Qualification **December 2026**

- Qualification for the American Invitational Mathematics Exam.

SKILLS

Technical Skills:

- C, C++, Java, Python proficiency
- GEANT4, GAIA DR3 experience
- Machine Learning experience

Languages: English (Native), Spanish (Proficient), ASL (Conversational)